

DA 3014 – Statistics Laboratory III – Assignment 3

Total: 100 marks

Submission Deadline: 11:59PM on 11th October, 2026

Instructions:

- Use **Python** for all analyses in this assignment.
- Compile your answers to all parts into a single report, including the necessary tables, figures, and other outputs. Convert the completed report to **PDF format** and name the file using your **Index Number**.
- Submit the PDF report, Python code and dataset (in CSV format) used for the analysis to the LMS by **11:59 PM on 11th October 2026**, all named with your Index Number.
- All analyses and reported results must be reproducible from the submitted Python code.

Project Overview

In this assignment, you are expected to compare a **binary logistic regression model** with **machine learning classification models** using a real-world dataset. You should critically evaluate the models in terms of their predictive performance, important predictors, and interpretability.

Part A [Finding an Appropriate Dataset]

5 marks

Choose a secondary dataset that meets the following criteria:

- It must be suitable for fitting binary classification models.
- It should contain multiple covariates, preferably 10–20 variables, with a reasonable combination of categorical and numerical variables.
- It must be obtained from a reliable source and must be a real-world dataset, not a synthetic dataset.
- It should contain more than 1,000 observations and preferably fewer than 500,000 observations.
- It should not be a commonly used dataset, such as the ‘Iris’, ‘Boston Housing’ or ‘Titanic’ datasets.
- A Google form is available for you to provide details of your selected dataset and reserve it for you, as soon as you have made your choice.
 - Link to the Google form: <https://forms.gle/rQswG7m61BDe9ZhF7>
- Each student must use a unique dataset. A Google Sheet is provided to keep track of selected datasets. Datasets will be reserved on a first-come, first-served basis.
 - Link to the Google Sheet:
<https://docs.google.com/spreadsheets/d/13oLQlSwbh2La4fjJ8CUQPhryxGgHhHQLMTusx3HDZPE/edit?usp=sharing>

In your report, clearly state the source, background, and a brief description of the dataset, including the response variable and covariates.

Part B [Exploratory Data Analysis (EDA)]**15 marks**

1. Perform the necessary data cleaning and preprocessing steps.
2. Examine the distributions of all variables using appropriate numerical summaries and visualisations.
3. Investigate the relationships between the covariates and the response using appropriate graphical summaries and statistical methods.
4. Did you observe any influential observations, outliers, or unusual patterns that may affect the modelling? Justify your answer.
5. Based on your EDA, identify two variables or relationships that you expect to be particularly important for prediction and explain why..
6. Split the data into training and test sets using an 80:20 ratio.

Part C [Traditional Statistical Model]**15 marks**

1. Fit a binary logistic regression model using all selected covariates to predict the binary outcome using the training data.
2. Identify and interpret the three most statistically significant predictors, including the direction and practical interpretation of their effects.
3. Obtain a reduced model using an appropriate feature-selection procedure.
4. Assess the adequacy and predictive performance of the final logistic regression model using appropriate measures.

Part D [Machine Learning Models]**20 marks**

1. Apply two machine learning classification methods (Random Forest + one other method) to the same prediction problem.
2. For each machine learning model:
 - a. Explain briefly how the method works.
 - b. Describe any additional preprocessing required.
 - c. Explain how model tuning was performed.
 - d. Evaluate the model's predictive performance on the test data.
 - e. Assess whether the model shows evidence of overfitting.
 - f. Identify the most important predictors using an appropriate variable-importance measure.

Part E [Discussion]**20 marks**

Answer the following questions, providing sufficient evidence from appropriate analyses to support your conclusions.

1. Do the machine learning models provide an advantage over logistic regression for your dataset? Justify your answer using appropriate performance measures.
2. If you were responsible for deploying one of the models in practice, which model would you recommend and why?
3. Were your initial expectations from the EDA regarding important predictors supported by the modelling results? Explain.

4. Does the Random Forest identify the same important predictors as the logistic regression model? Discuss any similarities or differences.
5. Can the best-performing model for your dataset be further improved by changing the classification threshold? Investigate and justify your conclusion.

Submission Details

- **Deadline:** 11:59PM on 11th October 2026
- **Late Submission Policy:** A deduction of 5% of the total marks per day will be applied to late submissions, up to a maximum of one week after the submission deadline. Submissions made more than one week after the deadline will not be accepted.
- **Files required and Naming:**
 - Report (in PDF format) – “24adaXXX_Report.pdf”
 - Python Code – “24adaXXX_Code.ipynb”
 - Dataset (in CSV format) – “24adaXXX_Data.csv”
- **Submission Portal:** Submit all three files together in a single submission through the LMS course page.

Evaluation

You will be evaluated based on:

- The quality and completeness of the submitted **report**.
- The reproducibility and clarity of the submitted **Python code**.
- Your ability to explain and defend your analysis during the **viva**.

Project Breakdown and Marks Allocation

Item	Description	Marks
• Part A	Finding an appropriate dataset	5
• Parts B to E	Addressing questions in Part B to E	70
• Python Code	Reproducible and clear code	5
• Viva	Q&A based on your entire analysis	20
		Total 100 marks

Important: You should be able to explain and justify all aspects of your analysis during the viva.